



Department of Electronic & Telecommunication Engineering
University of Moratuwa, Sri Lanka

BM4152 Biosignal Processing

Assignment 01

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1. Smoothing Filters

1.1. Moving Average MA(N) Filter

1.1.1 Preliminaries

The `ECG_template` signal (Figure 1) shows a clear, low-noise single PQRST complex sampled at $f_s = 500$ Hz with a small P wave bump near 0.12 s, a sharp QRS complex peaking at approximately 0.75 mV near 0.21 s, and a broader T wave near 0.44 s. Zero-mean Gaussian white noise scaled to a target SNR of 5 dB was added to produce `nECG` (Figure 2). Zero-mean Gaussian white noise was generated and scaled to obtain a target SNR of 5 dB. The measured power of the clean ECG was first calculated as

$$P_s = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n].$$

The target SNR was converted from dB to linear scale using

$$\text{SNR}_{\text{linear}} = 10^{\text{SNR}_{\text{dB}}/10}.$$

For a target SNR of 5 dB, the required noise power was therefore calculated as

$$P_n = \frac{P_s}{10^{5/10}}.$$

Since the generated noise has zero mean, its variance is equal to its power. Hence, the noise was generated with a standard deviation of

$$\sigma_n = \sqrt{P_n}.$$

The noisy ECG was then obtained by adding the scaled noise to the clean ECG:

$$nECG[n] = x[n] + w[n].$$

At this SNR the large-amplitude R-wave remains just visible above the noise, but the smaller P-wave and T-wave are almost entirely obscured.

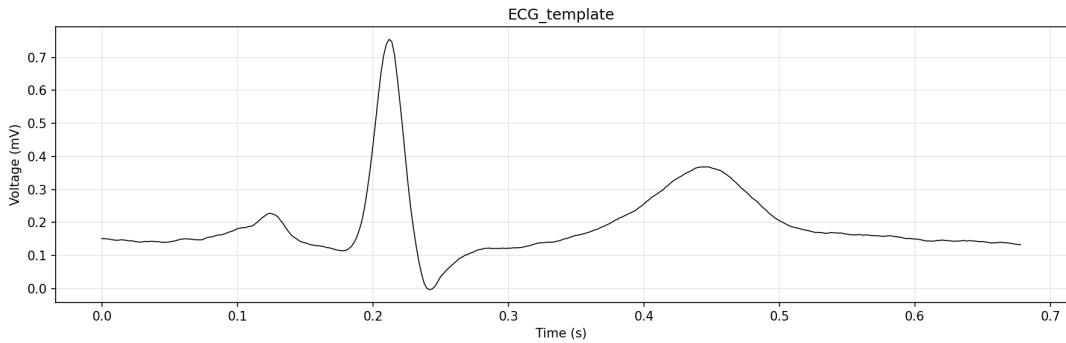


Figure 1: `ECG_template` signal.

The PSD of `nECG` (Figure 3) shows that most of the signal power is concentrated below approximately 20 Hz, where the main ECG components are present. At higher frequencies, the added white noise becomes more dominant.

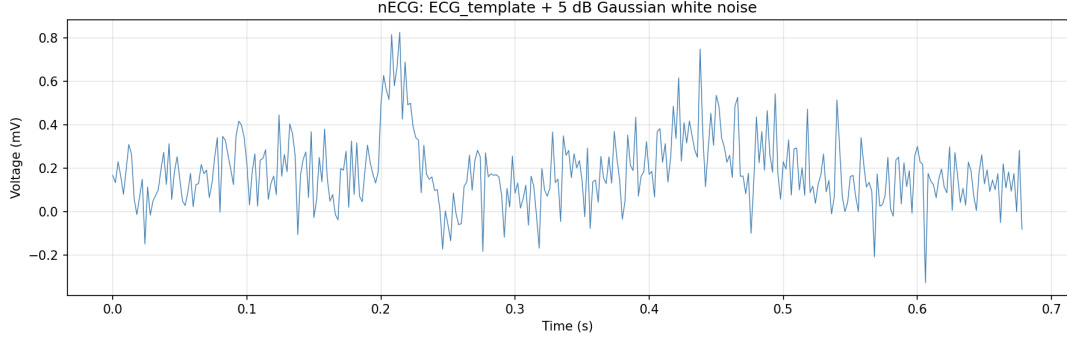


Figure 2: nECG: ECG_template with 5 dB Gaussian white noise added.

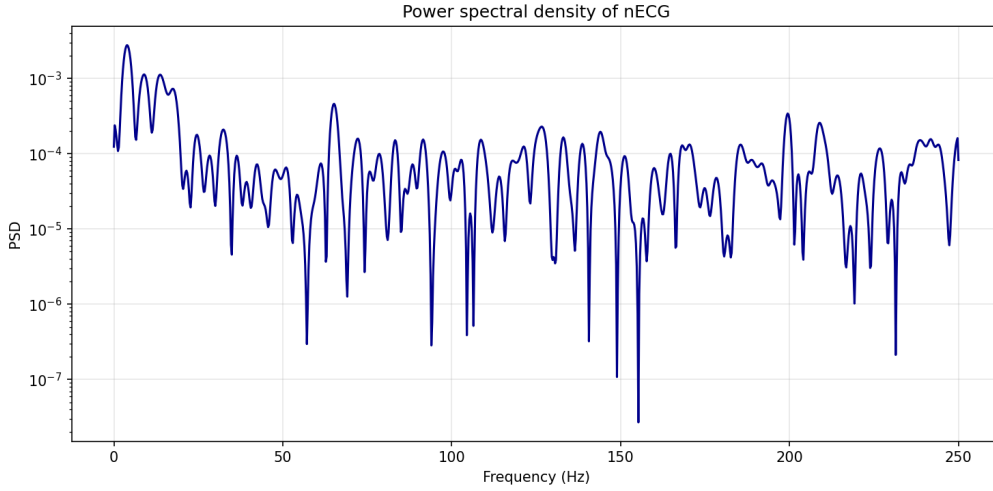


Figure 3: Power spectral density estimate of nECG.

1.1.2 MA(3) Filter: Custom Implementation

A custom function was used to implement the MA(3) filter directly using Equation (1):

$$y[n] = \frac{1}{3} \sum_{k=0}^2 x[n-k] = \frac{x[n] + x[n-1] + x[n-2]}{3}.$$

The MA(3) filter has a group delay of

$$D = \frac{N-1}{2} = \frac{3-1}{2} = 1 \text{ sample},$$

which corresponds to 2 ms at $f_s = 500$ Hz. After compensating for this delay, `ma3ECG_1` follows the main shape of the ECG `template`. The P-wave and T-wave are preserved, while the QRS peak is slightly reduced and smoothed. Some noise is still present because the 3-sample window provides only limited smoothing.

The overlapping PSDs in Figure 5 show the expected lowpass behaviour of the MA(3) filter. Below approximately 50 Hz, the filtered PSD is close to the noisy PSD, so the main ECG components are mostly preserved. Above approximately 50–60 Hz, the filtered PSD is lower than the noisy PSD, showing that the MA(3) filter reduces high-frequency noise.

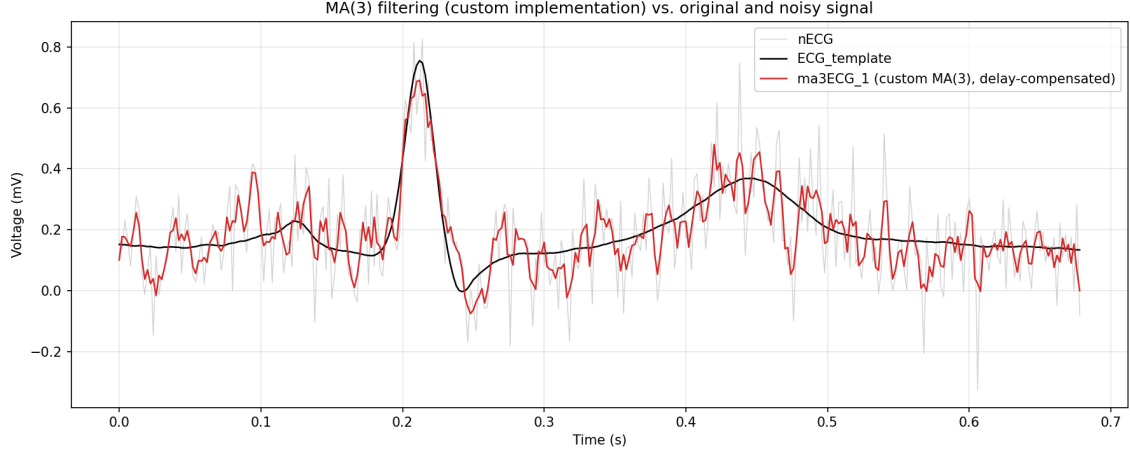


Figure 4: Delay-compensated ma3ECG_1 (custom MA(3) implementation) vs. ECG_template and nECG.

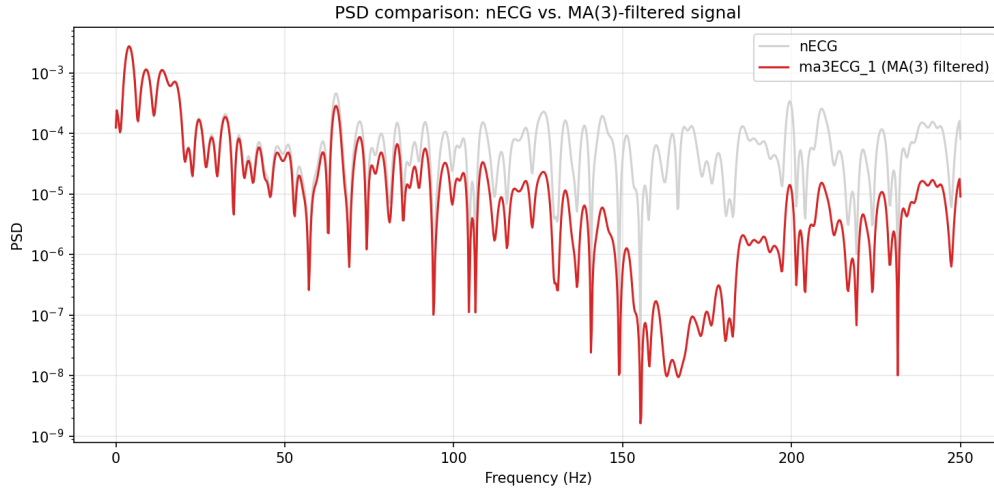


Figure 5: PSD comparison: nECG vs. ma3ECG_1 (MA(3) filtered).

1.1.3 MA(3) Filter: Built-in Function

The same MA(3) filter was implemented using `scipy.signal.lfilter()` with coefficients $b = [1/3, 1/3, 1/3]$, $a = [1]$, again compensating for the 1-sample group delay, giving ma3ECG_2. As expected, the resulting waveform (Figure 6) is visually identical to the custom implementation in Figure 4, confirming that the two implementations are equivalent.

Figure 7 shows the MA(3) filter's magnitude response, phase response, and pole-zero plot. The magnitude response is a smooth low-pass shape with a null at $f_s/3 \approx 167$ Hz. The phase response is approximately linear, with sudden jumps at frequencies where the magnitude response becomes zero, and the pole-zero plot shows two zeros symmetrically placed on the unit circle at $\pm 2\pi/3$ rad, with both poles located at the origin.

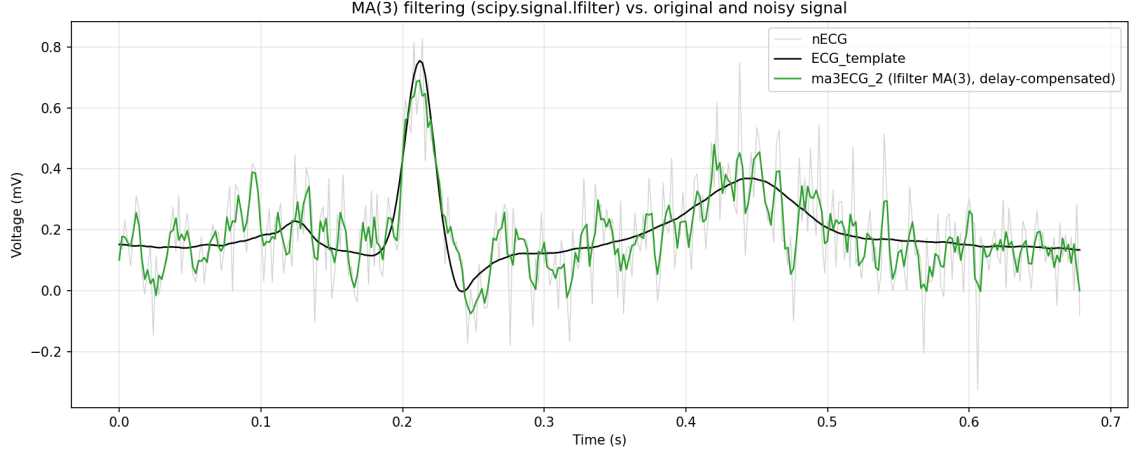


Figure 6: Delay-compensated `ma3ECG_2 (lfilter MA(3))` vs. `ECG_template` and `nECG`.

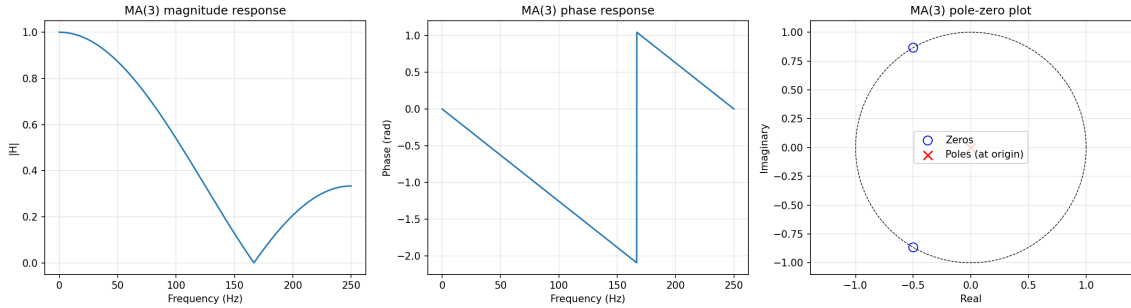


Figure 7: MA(3) filter: magnitude response, phase response, and pole-zero plot.

1.1.4 MA(10) Filter

Figure 8 compares the MA(3) and MA(10) filters. The MA(10) filter has a narrower main lobe, giving it a lower effective cutoff frequency and stronger attenuation of high-frequency components. However, it also has more ripples in the passband. Its zeros produce nulls at multiples of $f_s/10 = 50$ Hz.

The pole-zero plot shows that MA(10) has nine zeros evenly distributed around the unit circle, while MA(3) has only two zeros. The longer MA(10) filter also has a larger group delay:

$$D = \frac{N - 1}{2} = \frac{10 - 1}{2} = 4.5 \text{ samples},$$

compared with 1 sample for MA(3).

The time-domain comparison in Figure 9 shows that `ma10ECG` is smoother than `ma3ECG_2`, with less high-frequency noise. However, the QRS peak is also more reduced and rounded. This shows the trade-off between better noise reduction and preserving the original ECG shape when using a longer averaging window.

1.1.5 Optimum MA(N) Filter Order

The mean-squared-error (MSE) was used to measure the difference between the noise-free ECG signal and the filtered signal. It was calculated using

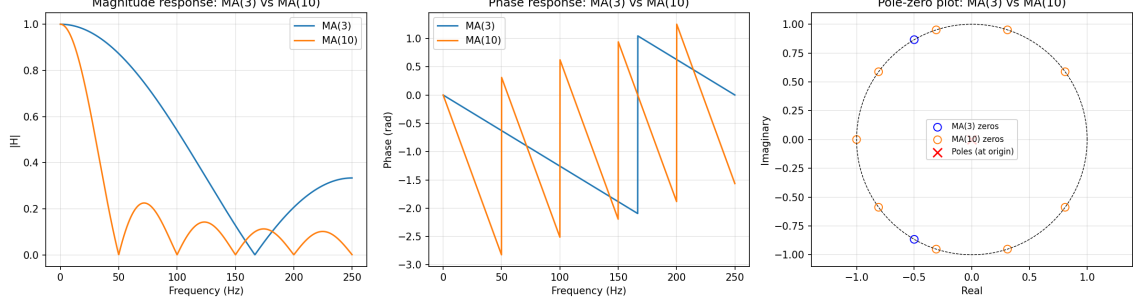


Figure 8: Magnitude response, phase response, and pole-zero plot: MA(3) vs. MA(10).

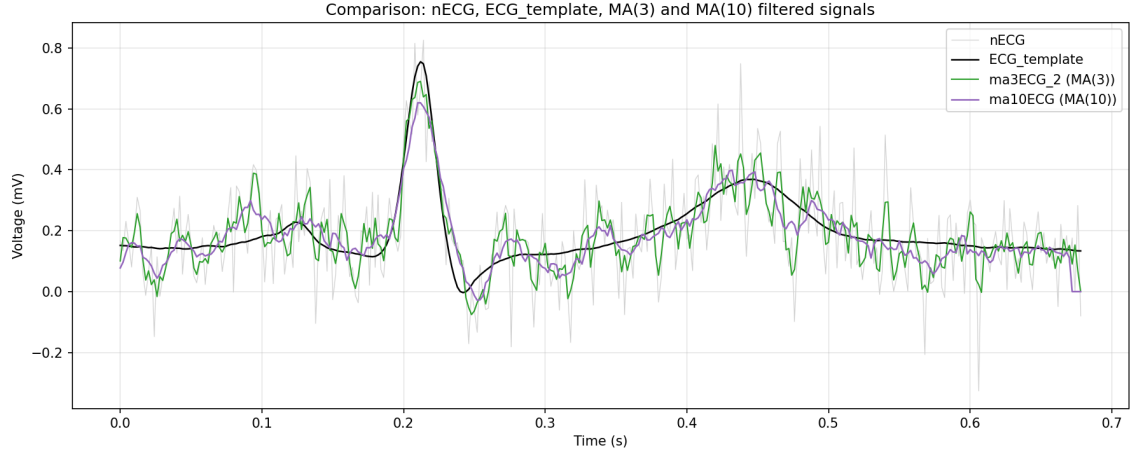


Figure 9: Comparison of nECG, ECG_template, ma3ECG_2 (MA(3)), and ma10ECG (MA(10)).

$$\text{MSE} = \frac{1}{M} \sum_{n=1}^M (x[n] - \hat{x}[n])^2,$$

where $x[n]$ is the noise-free `ECG_template`, $\hat{x}[n]$ is the delay-compensated output of the MA(N) filter, and M is the number of samples being compared.

The MA(N) filter was tested for $N = 1$ to $N = 100$. For each value of N , the MSE was calculated between the delay-compensated filtered signal and the noise-free `ECG_template`. As shown in Figure 10, the lowest MSE was obtained at $N = 12$, with an MSE of 0.0023. The MSE decreases quickly for small values of N and then gradually increases after $N = 12$.

Reasons for large MSE values at low and high filter orders

At low filter orders, the averaging window is short, so only a small number of samples are averaged together. As a result, the filter provides limited noise reduction, and much of the added white noise remains in the filtered ECG. This makes the filtered signal significantly different from the noise-free `ECG_template`, resulting in a large MSE. As the filter order increases, more samples are averaged and the noise is reduced, so the MSE decreases. However, at very high filter orders, the averaging window becomes too wide and starts to smooth not only the noise but also make changes in the ECG waveform. Therefore, although a higher filter order provides stronger noise suppression,

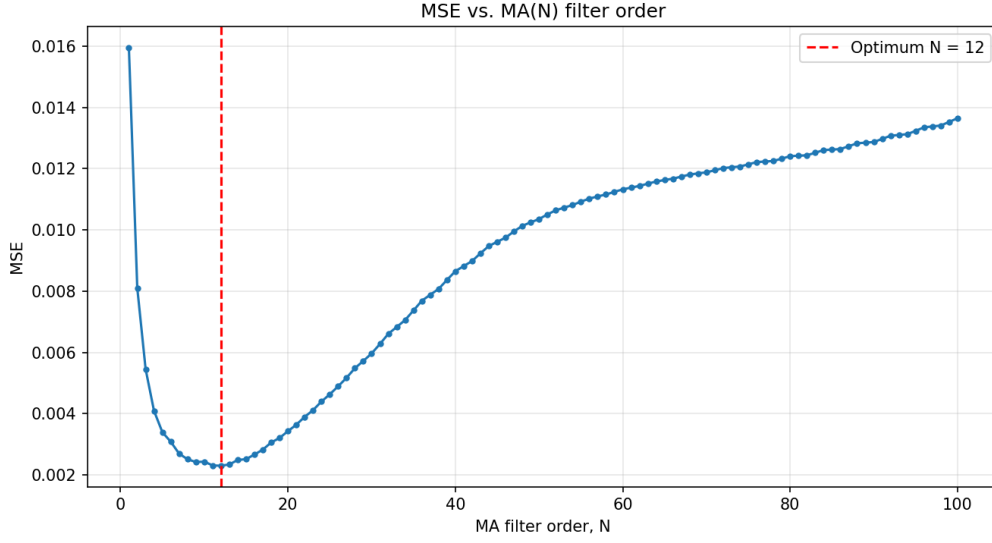


Figure 10: MSE vs. MA(N) filter order.

it also introduces greater distortion of the original ECG signal, causing the MSE to increase again.

1.2. Savitzky-Golay SG(N,L) Filter

1.2.1 Application of the SG(3,11) Filter

From the figure11, it can be observed that the SG(3,11) filter smooths the noisy signal while keeping the main shape of the ECG visible. The QRS complex is still clearly present, but some small fluctuations remain in the filtered signal. Larger variations can also be observed near the beginning and end of the signal, where the waveform is expected to be relatively flat.

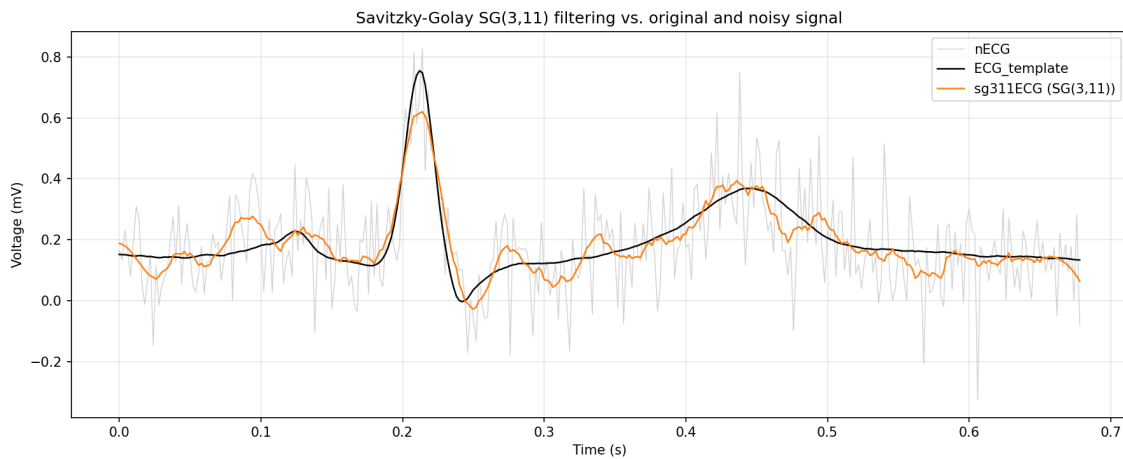


Figure 11: Savitzky-Golay SG(3,11) filtered signal (sg311ECG) vs. ECG_template and nECG.

1.2.2 Optimum SG(N,L) Filter Parameters

The polynomial order N (2 to 7) and half-window parameter L (2 to 25, giving an odd window length $L' = 2L + 1$ ranging from 5 to 51) were swept over a grid, computing the MSE against `ECG_template` for every valid combination (invalid combinations with $N > L' - 1$ were excluded). The resulting MSE surface (Figure 12) identifies the optimum parameters as $N = 2$, $L = 13$ (MSE = 0.001941).

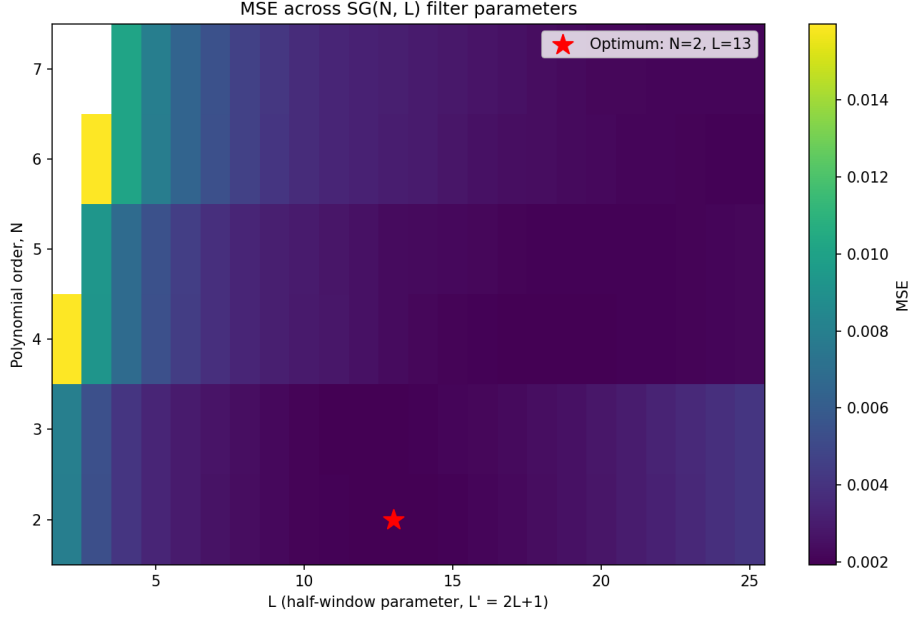


Figure 12: MSE across SG(N, L) filter parameters (pcolormesh view).

From Figure 13, it can be observed that the optimum SG(2,13) filter produces a smoother signal with less high-frequency noise than `sg311ECG`. However, the wider window also makes the QRS peak more rounded and slightly reduces its amplitude.

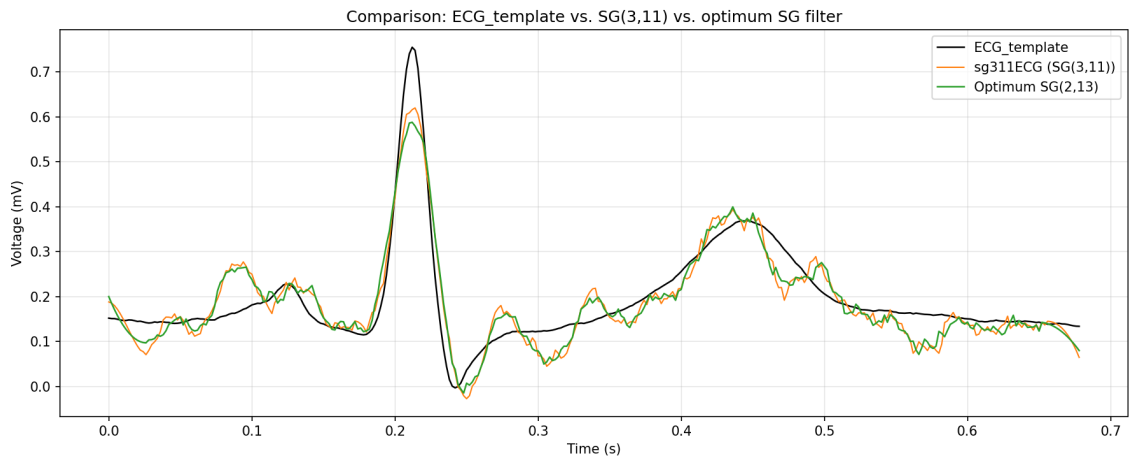


Figure 13: Comparison of `ECG_template`, `sg311ECG` (SG(3,11)), and the optimum SG(2,13) filter.

1.2.3 Comparison of Optimum MA(N) and SG(N,L) Filters

The optimum MA(12) filter (MSE = 0.0023) and the optimum SG(2,13) filter (MSE = 0.001941) are compared in Figure 14. The SG(2,13) filter gives a slightly lower MSE than the MA(12) filter. This is because the SG filter fits a polynomial to the samples within the window instead of simply taking their average. Therefore, it can preserve changes in the ECG waveform more effectively while still reducing noise. From the figure, both filtered signals have a similar overall shape, although the SG filter gives slightly smoother results. The downward change at the end of the MA(12) signal is caused by the delay compensation used for the causal MA filter and is not part of the actual filtered ECG signal.

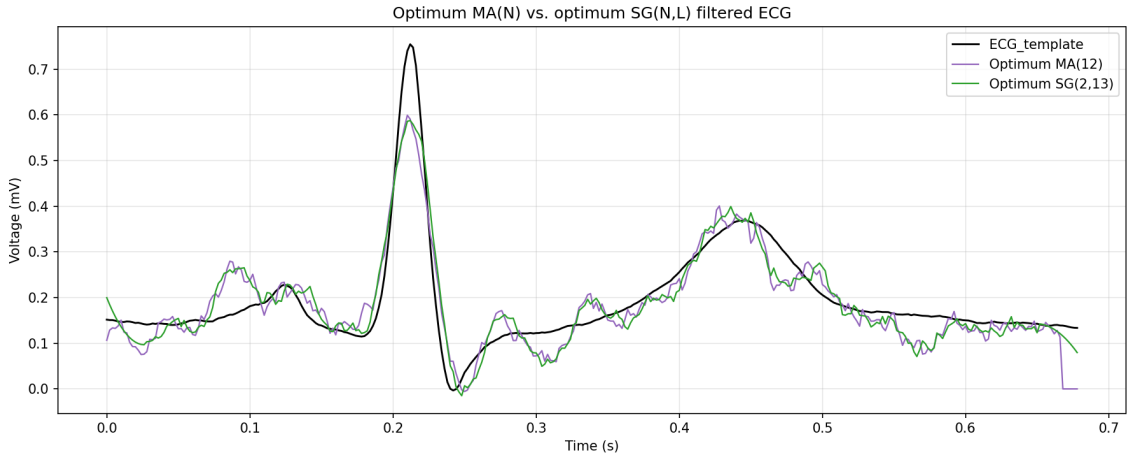


Figure 14: Optimum MA(12) vs. optimum SG(2,13) filtered ECG.

In terms of computational complexity, the MA(N) filter is simpler than the SG(N,L) filter. The MA filter mainly requires additions and a division for each output sample, while the SG filter uses a set of pre-computed coefficients to perform a longer FIR filtering operation. The SG filter also requires more computation when the coefficients are initially calculated because they are obtained using polynomial fitting. In this case, the optimum SG filter uses a 27-sample window, compared with 12 samples for the optimum MA filter. Therefore, the SG filter provides a slightly lower MSE, but the MA filter is simpler and computationally cheaper.

2. Ensemble Averaging

2.1. Signal with Multiple Measurements (ABR)

2.1.1 Preliminaries

Figure 15 shows the raw stimulus train together with the corresponding ABR+EEG recording. The stimulus train consists of repeated stimulus pulses occurring at regular intervals. The ABR recording is a much smaller-amplitude waveform compared with the stimulus. The ABR waveform is time-locked to the stimulus, meaning that similar ABR responses occur following each stimulus. However, the ABR is difficult to distinguish clearly from the background EEG/noise when the entire recording is viewed because of its relatively small amplitude.

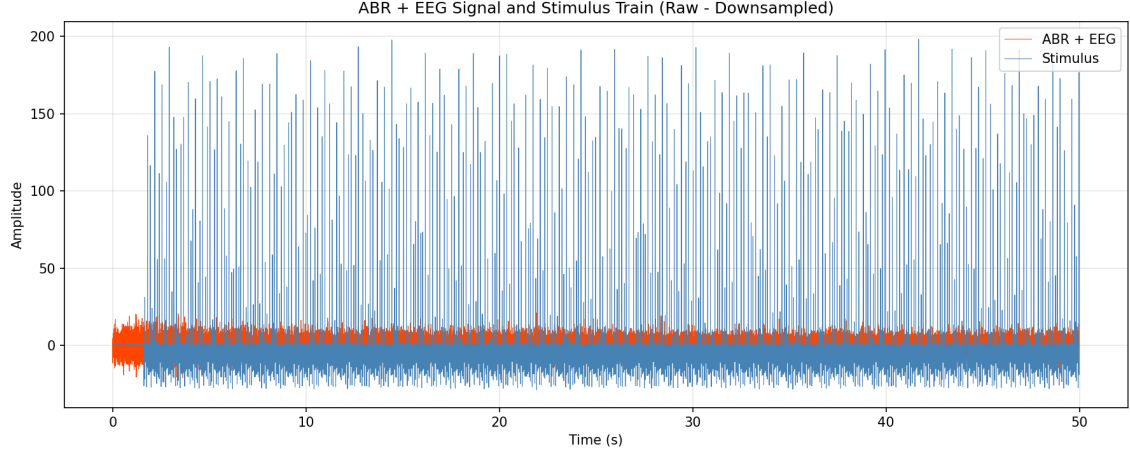


Figure 15: Raw stimulus train and ABR+EEG recording over the full acquisition period.

A voltage threshold of 50 was used to detect the beginning of each stimulus in the stimulus train. The leading edge of each detected stimulus was then identified to obtain a single starting point for each stimulus. In total, 1033 stimuli were detected. For each stimulus, an epoch was extracted from 2 ms before the stimulus onset to 10 ms after the stimulus onset. These epochs contain the corresponding ABR responses to each stimulus. The 1033 extracted ABR epochs were then averaged together using ensemble averaging. Since the ABR response occurs at approximately the same time after each stimulus, the repeated ABR components are preserved during averaging, while random background noise is reduced. Although the ABR waveform is difficult to distinguish from noise in an individual recording, the ensemble-averaged waveform clearly reveals the characteristic ABR morphology, including several distinct peaks known as waves I–VI. The resulting ensemble-averaged ABR waveform is shown in Figure 16.

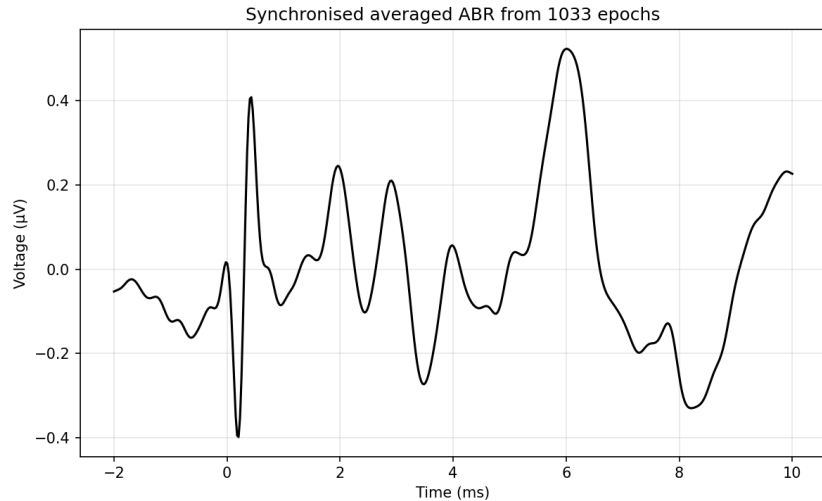


Figure 16: Ensemble-averaged ABR waveform from 1033 epochs, showing resolvable waves I–VI.

2.1.2 Improvement of the SNR

Figure 17 shows the progressive MSE, MSE_k , calculated between the ensemble-averaged ABR template and the progressive average of the first k epochs, where k ranges from 1 to 1033. The MSE decreases rapidly as the number of averaged epochs increases. A large reduction in MSE can be observed during the first approximately 100 epochs, after which the reduction becomes more gradual. This indicates that the progressive average becomes increasingly similar to the final ensemble-averaged ABR as more epochs are included.

The reduction in MSE is related to the improvement in SNR obtained through ensemble averaging. Since the ABR response is repeated at approximately the same time for each stimulus, the ABR signal is preserved during averaging, while the random noise is reduced. Theoretically, averaging k independent noise realisations reduces the noise amplitude by a factor of $1/\sqrt{k}$, resulting in an improvement in SNR as the number of averaged epochs increases. Therefore, the rapid decrease in MSE at the beginning indicates that most of the improvement occurs when the first set of epochs is averaged, while additional epochs provide progressively smaller improvements. At $k = 1033$, the progressive average becomes the final ensemble average, resulting in the MSE approaching zero.

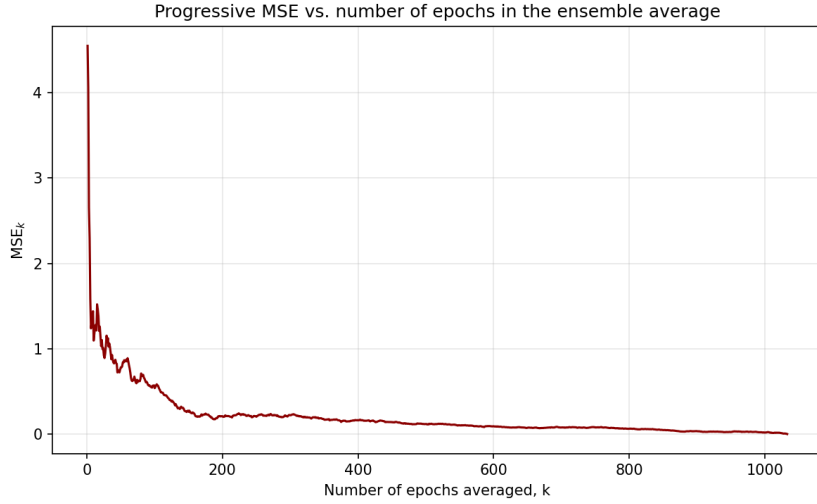


Figure 17: Progressive MSE, MSE_k , versus the number of ABR epochs k included in the running average.

2.2. Signal with Repetitive Patterns (ECG)

2.2.1 Viewing the Signal and Addition of Gaussian White Noise

The raw recording recording (Figure 19) exhibits clearly defined, regularly-spaced QRS complexes with a consistent morphology across the 60-second recording, indicating a stable underlying rhythm. A zoomed-in version is also included to view the ECG features more clearly. A single representative PQRST waveform was extracted and stored as `ECG_template` (Figure 20), showing well-defined P, Q, R, S and T waves.

Zero-mean Gaussian white noise scaled to a target SNR of 5 dB was added to the ECG recording to produce `nECG` (Figure 21). At this SNR, the high-amplitude R-waves remain visible above the noise floor, but the lower-amplitude P and T waves and fine detail

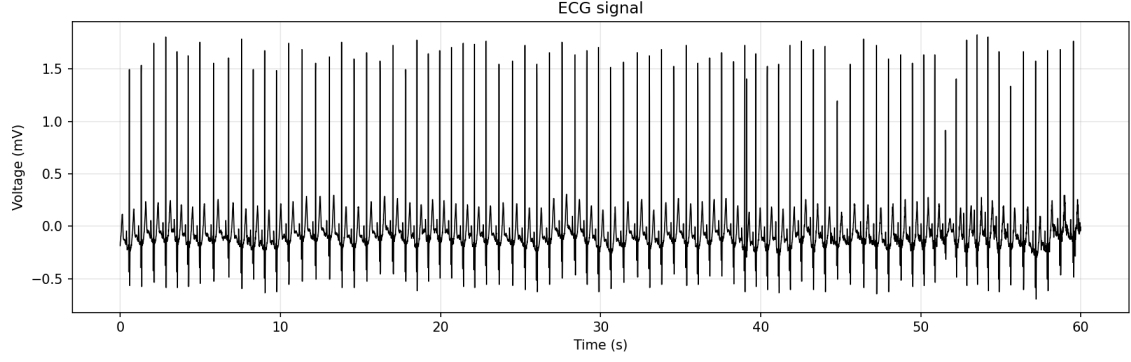


Figure 18: Raw ECG recording (Lead II).

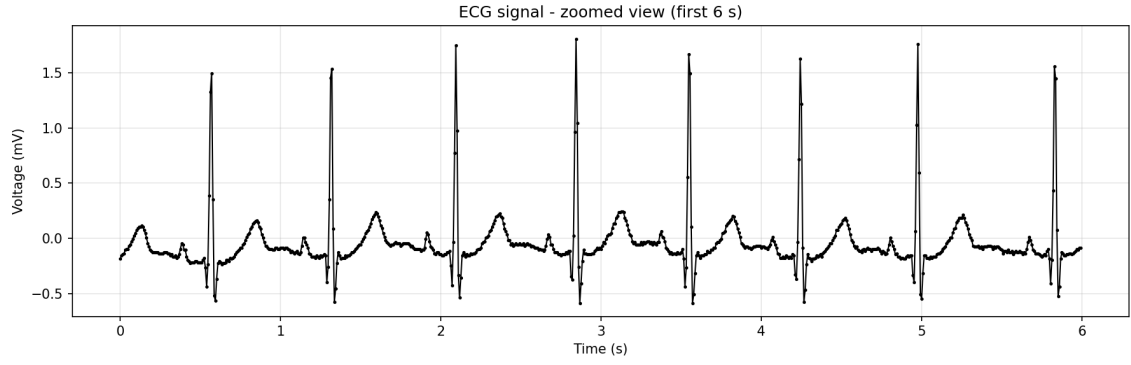


Figure 19: Raw ECG recording (Zoomed in version).

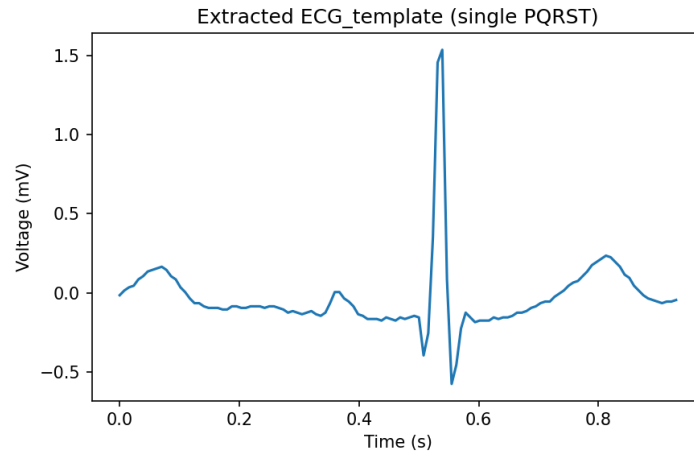


Figure 20: Extracted single PQRST waveform used as `ECG_template`.

within the ST segment are largely obscured, motivating the need for a noise reduction technique such as ensemble averaging that specifically exploits the periodic, repetitive structure of the ECG waveform.

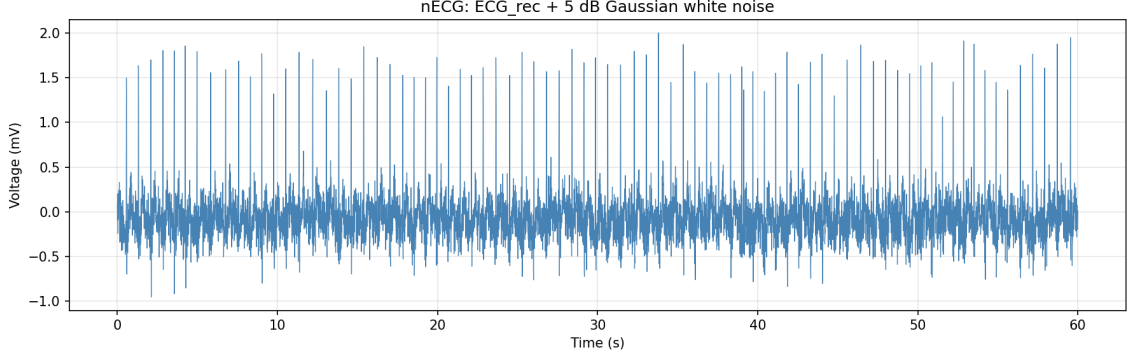


Figure 21: Noisy ECG signal (**nECG**) after adding 5 dB Gaussian white noise.

2.2.2 Segmenting ECG into Separate Epochs and Ensemble Averaging

The normalised cross-correlation between **ECG_template** and **nECG** (Figure 22) shows a series of sharp and regularly-spaced peaks, corresponding to the individual heartbeats. The peaks are clearly distinguishable from the surrounding noise, although the added noise reduces the maximum correlation value to approximately 0.113.

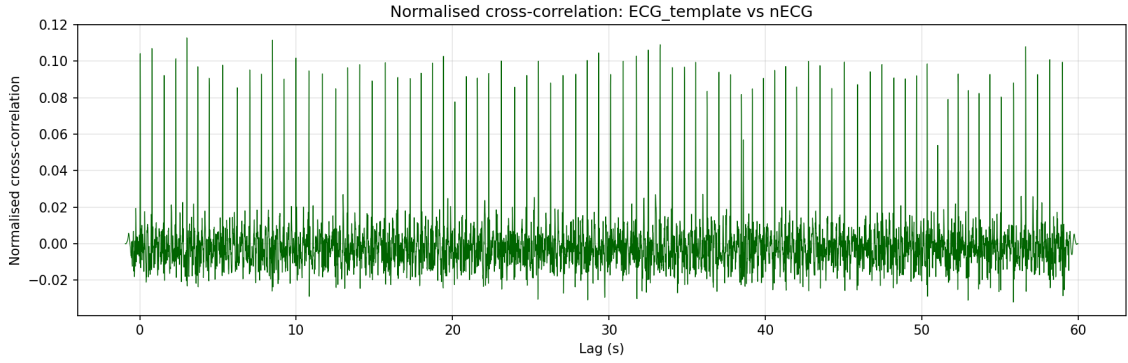


Figure 22: Normalised cross-correlation between **ECG_template** and **nECG**.

A relative threshold of 40% of the maximum correlation value (≈ 0.045) was therefore used (via `scipy.signal.find_peaks`) rather than a fixed absolute threshold, since the absolute correlation magnitude is data- and noise-dependent. This detected 79 ECG pulses over the 60-second recording.

Figure 23 shows five individual segmented pulses, sampled evenly across the recording, both overlaid and stacked. The overlaid view confirms that the cross-correlation-based segmentation correctly aligns the dominant R-wave across epochs, with only minor residual jitter, while the stacked view illustrates that the underlying noise level and fine waveform detail (P- and T-wave visibility) vary noticeably from beat to beat.

The progressive MSE for the ECG ensemble average (Figure 24) shows a similar trend to the ABR. The MSE decreases quickly during the first ~ 10 pulses, from approximately 0.16 to 0.04. After this, the MSE decreases more slowly as more pulses are added. This shows that ensemble averaging reduces the noise more effectively when the number of averaged pulses is increased, following the expected $1/\sqrt{k}$ noise-suppression behaviour.

Figure 25 compares a single noisy ECG pulse with ensemble-averaged ECG pulses obtained using 10 pulses and all 79 segmented pulses. The single noisy pulse contains a

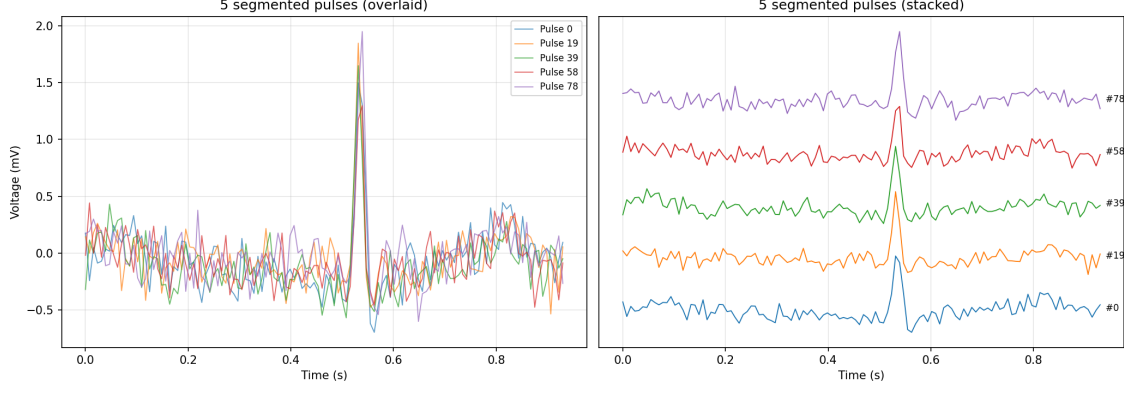


Figure 23: Five representative segmented ECG pulses shown overlaid (left) and stacked (right), sampled evenly across the recording.

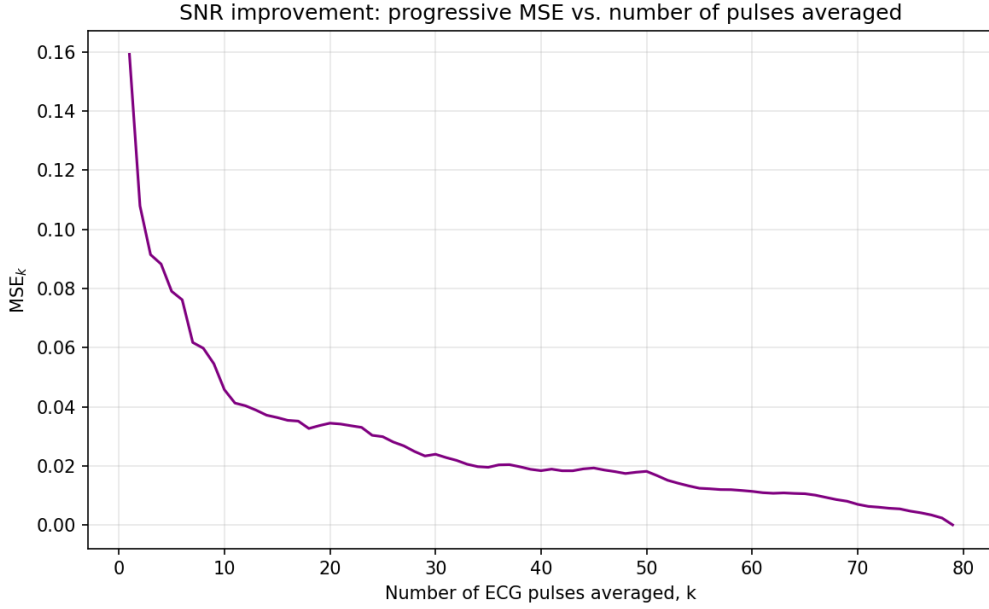


Figure 24: Progressive MSE, MSE_k , versus the number of ECG pulses k included in the running ensemble average.

noticeable amount of random noise, particularly in the low-amplitude regions where the noise can be comparable to the ECG signal. After averaging 10 pulses, the random noise is reduced and the ECG waveform becomes clearer. Averaging all 79 pulses provides further noise reduction and produces a smoother waveform. The R-wave is clearly visible even in the noisy pulse because its amplitude is much larger than the noise.

Justification: Correlation-based segmentation vs R-wave detection. Using the maximum cross-correlation with the full PQRST template is preferable to using only R-wave detection to segment the ECG pulse train. Cross-correlation considers the shape of the entire ECG waveform, including the P, Q, R, S and T waves, rather than relying on only the R-wave. This allows the ECG pulses to be better aligned, even when there are small variations in the timing or shape of individual waves.

Correlation-based segmentation is also more robust to noise. A noisy ECG signal

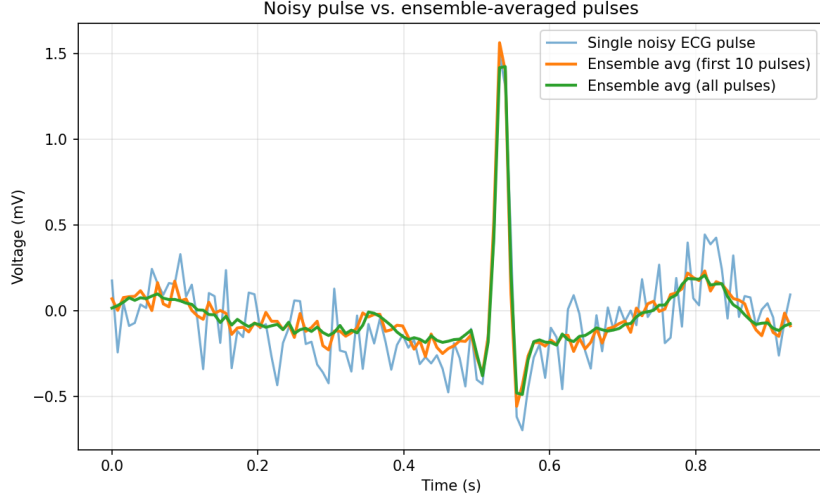


Figure 25: Comparison of a single noisy ECG pulse against ensemble averages of 10 and 79 pulses.

may contain noise peaks that could be incorrectly detected as R-waves, or noise may reduce the amplitude of a true R-wave and make it difficult to detect. In contrast, cross-correlation compares the complete waveform with the template, so a true ECG pulse can still produce a strong correlation even when some parts of the waveform are affected by noise. Therefore, using the maximum cross-correlation provides more reliable segmentation and better alignment of the ECG pulses, resulting in a clearer ensemble-averaged waveform.

3. Designing FIR Filters Using the Windowing Method

3.1. Characteristics of Window Functions

3.1.1 Rectangular Window: Effect of Length M

Figure 26 shows the impulse responses of FIR filters designed using a rectangular window for $M = 5, 50, 100$. As M increases, the impulse response becomes longer because the filter has more coefficients. The response for $M = 5$ is short, while $M = 50$ and $M = 100$ have increasingly longer responses. Therefore, increasing the window length increases the length of the FIR filter and allows a narrower transition band, but also increases the number of coefficients and the computational cost.

The corresponding lowpass filter response ($\omega_c = 0.4\pi$) is shown in Figure 27. As M increases from 5 to 50 and 100, the transition band becomes narrower and the filter becomes sharper. Therefore, a larger window length gives better frequency selectivity. However, increasing M does not significantly reduce the ripple near the cutoff frequency. Instead, the ripples become narrower and more closely spaced. This ripple is caused by the abrupt edges of the rectangular window.

The phase response is approximately linear for all three filters. This means that the different frequency components experience a constant time delay. As M increases, the delay also increases, given by $(M - 1)/2$ samples. The small irregularities in the phase response in the stopband occur near frequencies where the magnitude is very close to zero.

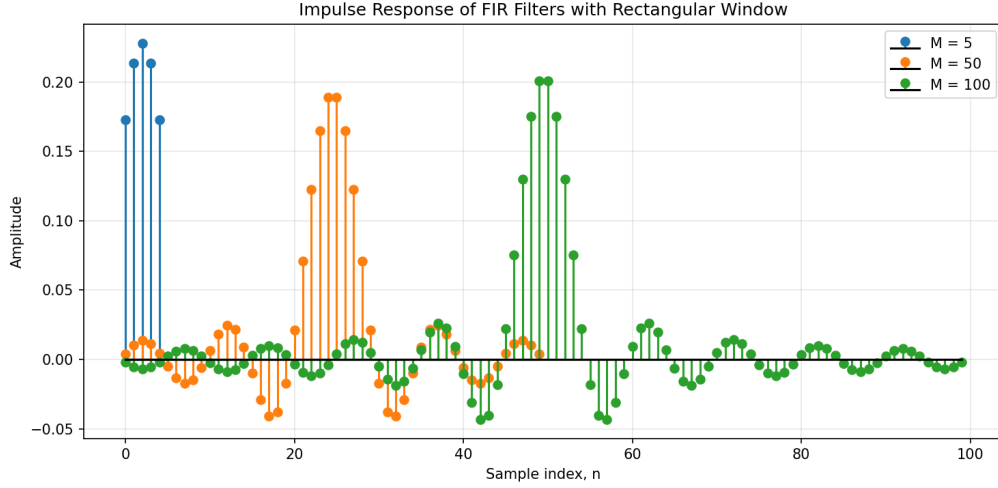


Figure 26: Rectangular window impulse response, overlaid for $M = 5, 50, 100$.

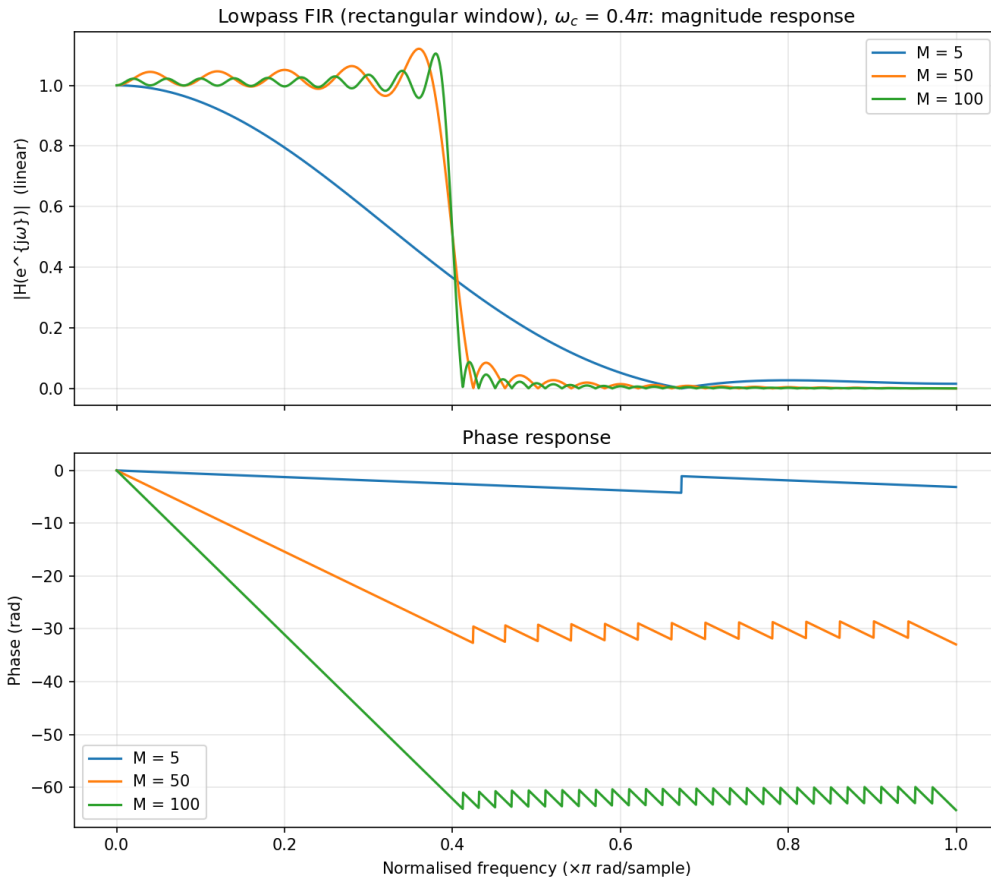


Figure 27: Lowpass FIR filter ($\omega_c = 0.4\pi$) built with a rectangular window: magnitude response (top) and phase response (bottom), for $M = 5, 50, 100$.

3.1.2 Comparison of Rectangular, Hanning, Hamming and Blackman Windows ($M = 50$)

Figure 28 shows the four windows in the time domain. The rectangular window is flat with an abrupt discontinuity at both edges. The Hanning and Hamming windows are

both raised-cosine shapes, but the Hamming window does not taper fully to zero at its edges (it settles at approximately 0.08 rather than 0), which leaves a small residual discontinuity. The Blackman window is the smoothest of the three tapered windows, incorporating an additional cosine term that brings its edges essentially to zero, at the cost of a visibly narrower effective main-lobe width relative to its total span (a more gradual taper concentrated nearer the centre).

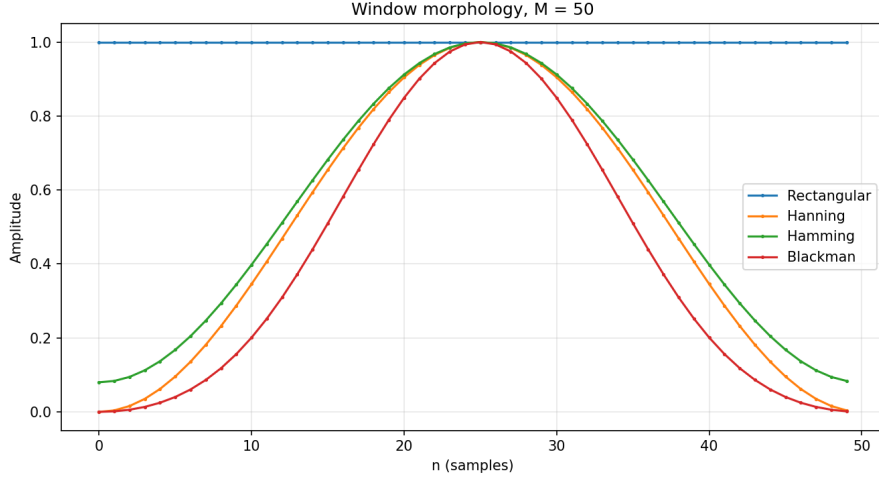


Figure 28: Window morphology for the rectangular, Hanning, Hamming and Blackman windows, $M = 50$.

The corresponding magnitude responses (Figure 29) illustrate the classic transition-width vs. sidelobe-attenuation trade-off. On the linear-scale plot, the rectangular window shows the largest passband ripple (peaking near 1.12), while the Hanning, Hamming and Blackman windows are nearly flat and ripple-free approaching the cutoff. On the log-scale plot this trade-off is much clearer. The rectangular window's sidelobes decay very slowly, remaining within roughly -30 to -45 dB over many lobes; the Hanning window's sidelobes fall faster, reaching below -80 dB; the Hamming window's sidelobes are lower still, dropping below -90 dB with a markedly deep null structure; and the Blackman window gives the lowest sidelobes of all, decaying to the plotted floor of -120 dB within a few lobes. This comes at the cost of transition sharpness. All three smooth windows exhibit a visibly wider transition band than the rectangular window at the same M , since their tapering reduces spectral leakage at the expense of main-lobe width.

Figure 30 confirms that all four windowed filters retain a linear phase response within the passband (the straight-line segment up to $\approx 0.4\pi$), since every window used here is symmetric about its centre sample, guaranteeing a constant group delay of $(M - 1)/2 = 24.5$ samples in each case. The vertical offsets between the curves beyond the cutoff frequency are caused by the different locations of the zeros of each filter, which can produce phase changes near the frequencies where the magnitude response becomes zero.

3.2. FIR Filter Design and Application Using the Kaiser Window

3.2.1 Signal Inspection

The raw ECG with noise signal is shown in Figure 31. The signal contains the main ECG waveform along with low-frequency baseline variation and high-frequency noise.

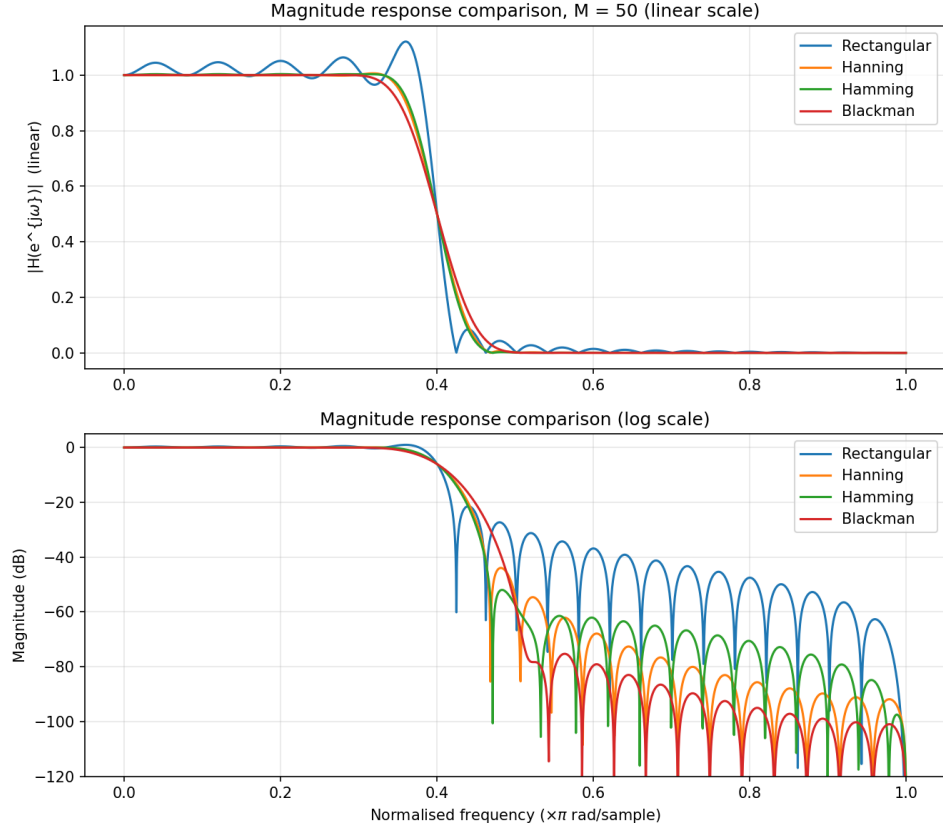


Figure 29: Magnitude response comparison for the four windows, $M = 50$: linear scale (top) and log (dB) scale (bottom).

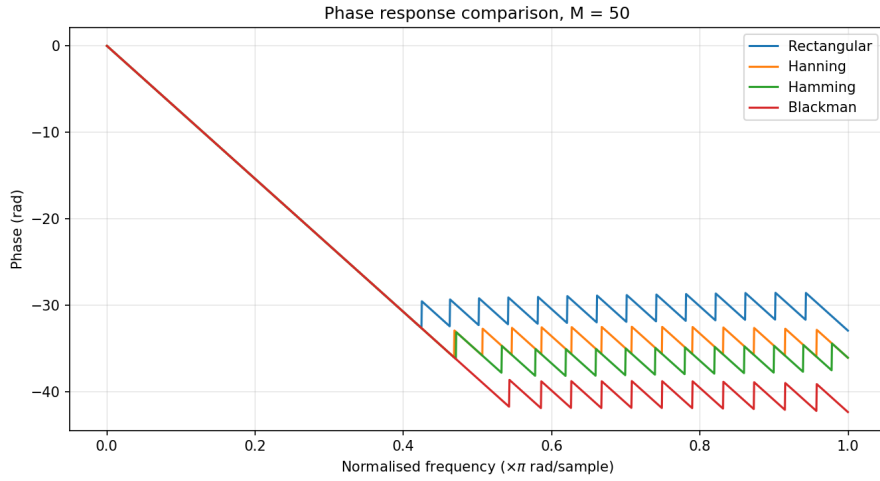


Figure 30: Phase response comparison for the four windows, $M = 50$.

The PSD of the noisy ECG is shown in Figure 32. Most of the ECG energy is concentrated in the low-frequency range, which is consistent with the typical bandwidth of a normal ECG. The peaks at 50, 100 and 150 Hz are due to powerline interference and its harmonics, while the higher-frequency components are mainly due to noise.

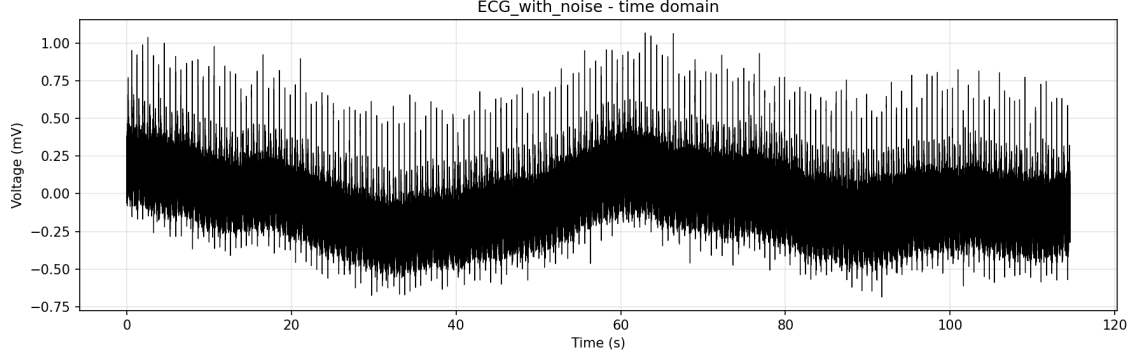


Figure 31: Time-domain ECG with noise recording.

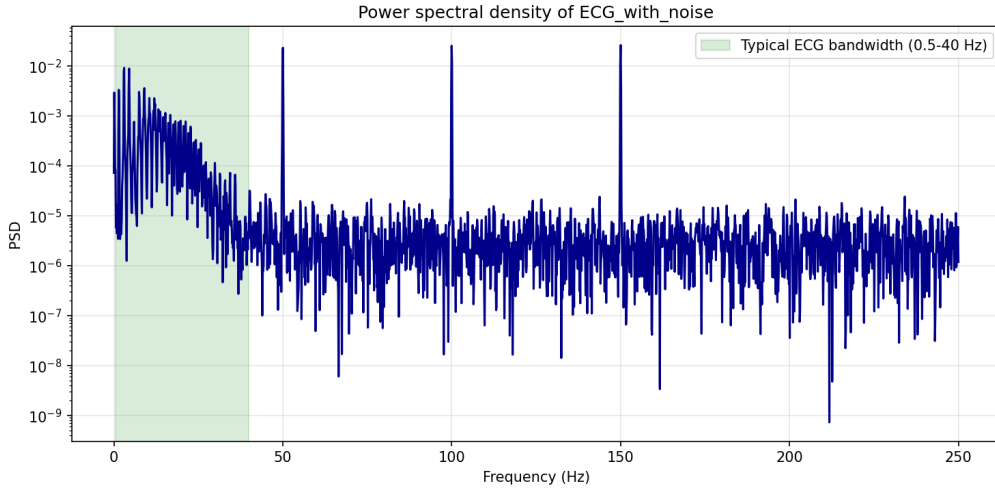


Figure 32: Power spectral density estimate of ECG with noise, with the typical diagnostic ECG bandwidth (0.5-40 Hz) shaded.

3.2.2 Filter Parameters and Kaiser Window Design

Based on the PSD observations above, three FIR filters were designed using the Kaiser window method via `scipy.signal.kaiserord()` and `scipy.signal.firwin()`:

- **Highpass filter:** stopband edge $f_{stop} = 3$ Hz, passband edge $f_{pass} = 5$ Hz and $\delta = 0.01$.
- **Lowpass filter:** passband edge $f_{pass} = 125$ Hz, stopband edge $f_{stop} = 135$ Hz, and $\delta = 0.01$.
- **Notch/comb filter:** three narrow stopbands centred at $f_{stop1} = 50$ Hz, $f_{stop2} = 100$ Hz and $f_{stop3} = 150$ Hz (each ± 2 Hz wide).

A common target attenuation of $A = 40$ dB was used for the filters. The Kaiser window parameters were calculated using

$$M \approx \frac{A - 8}{2.285 \Delta\omega}, \quad \Delta\omega = 2\pi \frac{\Delta f}{f_s}$$

and

$$\beta = \begin{cases} 0.1102(A - 8.7), & A > 50, \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21), & 21 \leq A \leq 50, \\ 0, & A < 21. \end{cases}$$

For $A = 40$ dB, this gives $\beta \approx 3.40$. The corresponding filter parameters obtained using `scipy.signal.kaiserord()` are:

Filter	Transition width	M	β
Highpass	2 Hz (3–5 Hz)	561	3.40
Lowpass	10 Hz (125–135 Hz)	113	3.40
Notch/comb	1 Hz (per band edge)	1119	3.40

3.2.3 Window and Response Visualisation

Figure 33 shows the Kaiser window, magnitude response, and phase response of the three filters. The Kaiser windows have a smooth shape and are adjusted to the different filter lengths.

The highpass filter strongly attenuates frequencies below approximately 5 Hz while allowing higher frequencies to pass. The lowpass filter allows frequencies up to approximately 125 Hz and attenuates higher frequencies. The notch/comb filter produces three narrow rejection bands at 50, 100 and 150 Hz to remove the powerline interference.

The phase responses of all three filters are approximately linear, which indicates that they have a constant group delay. The notch/comb filter has the largest delay because it has the largest filter length. For $M = 1119$, the group delay is

$$D = \frac{M - 1}{2} = \frac{1119 - 1}{2} = 559 \text{ samples.}$$

At $f_s = 500$ Hz, this corresponds to

$$D = \frac{559}{500} = 1.118 \text{ s.}$$

3.2.4 Sequential Filtering with Group Delay Compensation

Each filter was applied sequentially using `scipy.signal.lfilter()`, with the constant group delay of $(M - 1)/2$ samples. Figure 34 shows the progressive effect of each stage. The highpass stage successfully removes the slow baseline wander observed in the raw signal, flattening the envelope around zero. The lowpass stage produces a modest further smoothing, primarily suppressing the very highest-frequency components. The final notch stage visibly sharpens and cleans the individual QRS-like peaks compared to the earlier stages.

3.2.5 Combined Response and Final PSD

The combined magnitude response of the cascaded filter (obtained by convolving the three impulse responses, equivalent to applying all three filters in series) is shown in Figure 35. It clearly shows a composite passband between approximately 5 and 125 Hz, with three sharp notches at 50, 100 and 150 Hz and strong attenuation outside the 5–125 Hz band, exactly matching the intended design.

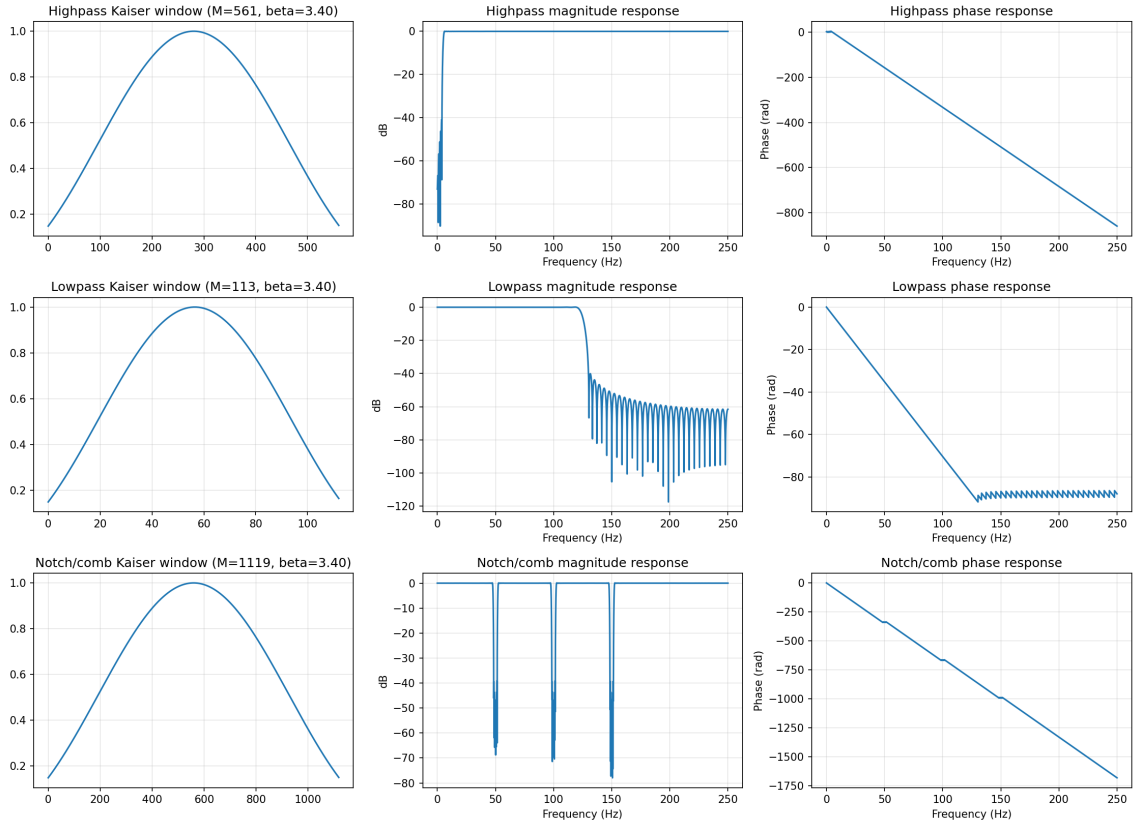


Figure 33: Kaiser window shape (left), magnitude response (middle) and phase response (right) for the highpass, lowpass, and notch filters.

Figure 41 compares the PSD of the original noisy ECG against the final filtered signal. The three powerline peaks at 50, 100 and 150 Hz are suppressed heavily in magnitude, confirming effective removal of the narrowband interference. The filter also reduces unwanted frequencies below 5 Hz and above 125 Hz. Most importantly, the ECG frequencies in the 5–40 Hz range are largely preserved. This means the filter successfully removes the main noise while keeping the important ECG signal.

4. IIR Filters

4.1. Realizing IIR Filters

4.1.1 Butterworth Lowpass Filter

A 4th-order Butterworth lowpass filter was designed using `scipy.signal.butter()` with a cutoff frequency of 125 Hz, which is the same cutoff frequency used for the FIR lowpass filter in Section 3.2.

The filter was designed using

$$[b, a] = \text{butter}(4, 125, \text{btype}='low', \text{fs} = 500).$$

The resulting numerator coefficients were

$$b = [0.09398085, 0.37592341, 0.56388511, 0.37592341, 0.09398085]$$

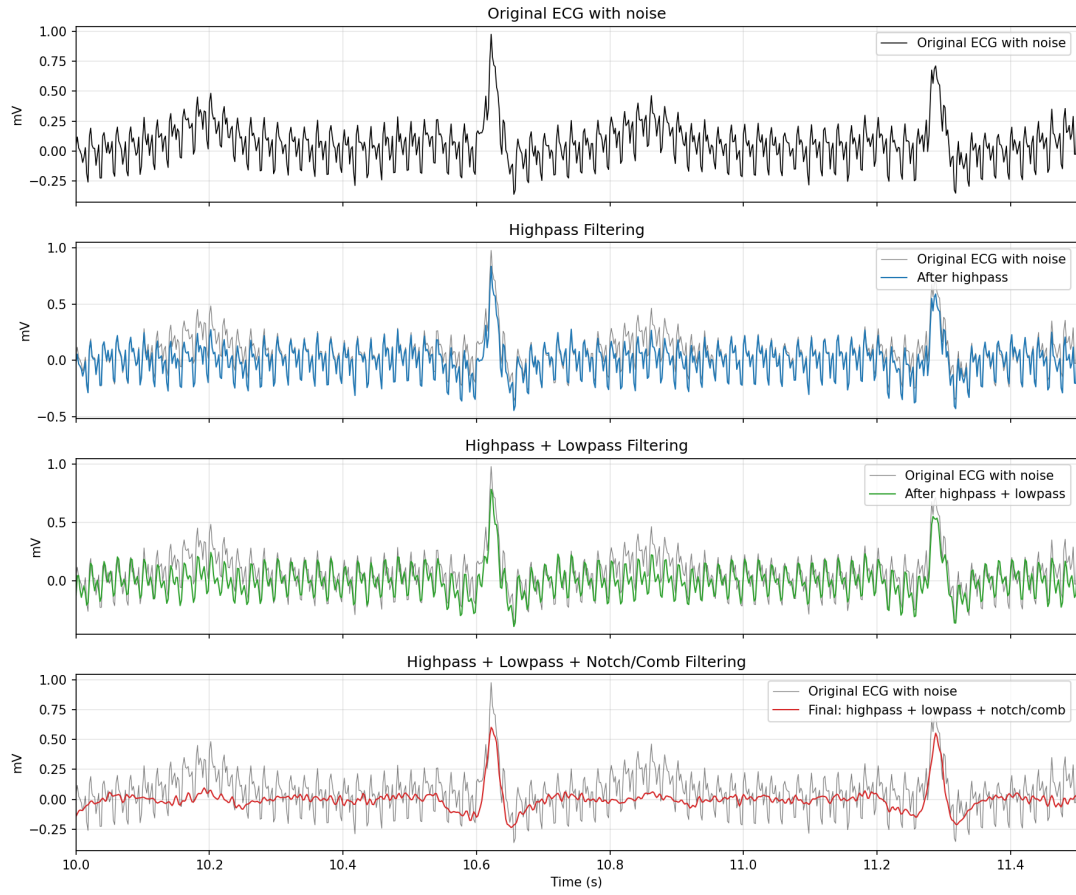


Figure 34: Time-domain effect of applying the highpass, lowpass and notch/comb filters sequentially, with group delay compensation at each stage (Zoomed in for clear visualisation).

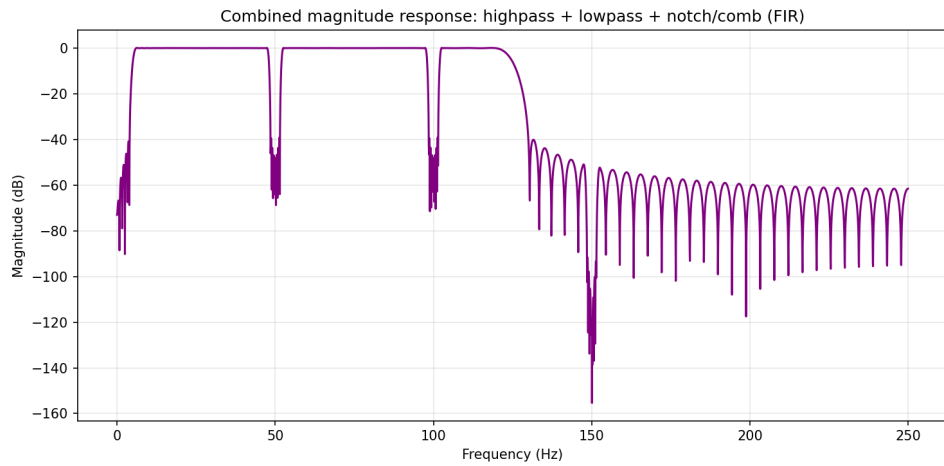


Figure 35: Combined magnitude response of the cascaded highpass, lowpass and notch/comb FIR filters.

and the denominator coefficients were

$$a = [1.00000000, -2.49800181 \times 10^{-16}, 0.486028822, -6.63518679 \times 10^{-17}, 0.0176648009].$$

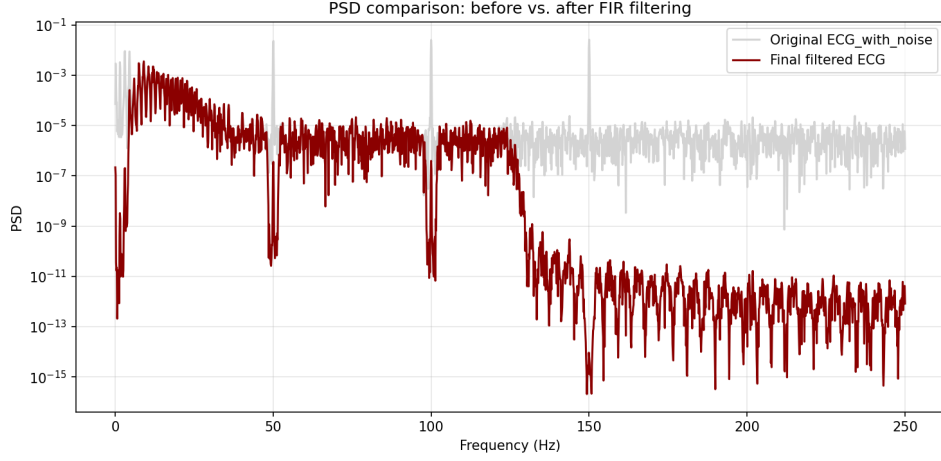


Figure 36: PSD comparison between the original `ECG_with_noise` signal and the final filtered ECG.

The very small values close to zero in the denominator coefficients are numerical rounding effects and can effectively be considered zero.

It should be noted that the full FIR attenuation specification was not used for the IIR design. An initial attempt to match the FIR specifications using `scipy.signal.buttord()` resulted in a very high order of $N = 42$. This caused numerical problems when using the direct-form (b, a) coefficients. Therefore, a fixed order of $N = 4$ was used. This gives a simpler and numerically stable IIR filter for comparison.

Figure 37 shows the magnitude, phase and group delay responses. The Butterworth filter has a smooth magnitude response without ripple. The phase response is nonlinear, and the group delay changes with frequency. The group delay increases around the cutoff frequency and then decreases at higher frequencies. This is different from the FIR filter, which has a constant group delay because it has a linear phase response.

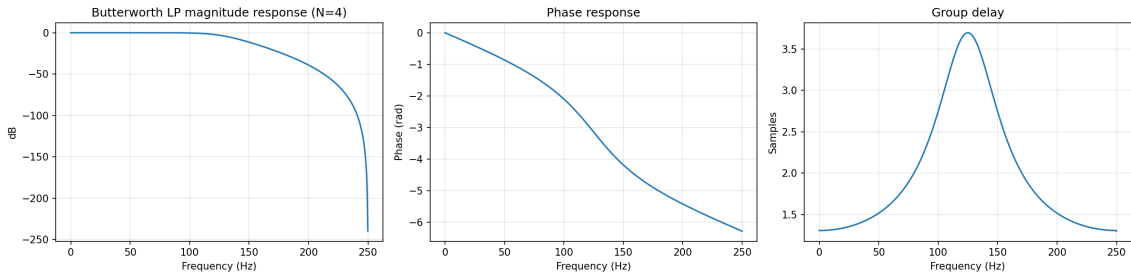


Figure 37: Butterworth lowpass filter ($N = 4$, cutoff 125 Hz): magnitude response (left), phase response (middle), and group delay (right).

4.1.2 Highpass and Comb Filters

Figure 38 shows the magnitude, phase and group delay responses of the Butterworth highpass filter and the IIR comb filter. The Butterworth highpass filter has order $N = 4$ and a cutoff frequency of 5 Hz. The filter coefficients obtained using `scipy.signal.butter()` are

$$b = [0.92117099, -3.68468397, 5.52702596, -3.68468397, 0.92117099]$$

and

$$a = [1.00000000, -3.83582554, 5.52081914, -3.53353522, 0.84855600].$$

The comb filter was designed using `scipy.signal.iircomb()` with a fundamental frequency of 50 Hz and $Q = 12.5$. The resulting coefficients are

$$b = [0.88783976, 0, 0, 0, 0, 0, 0, 0, 0, -0.88783976]$$

and

$$a = [1.00000000, 0, 0, 0, 0, 0, 0, 0, 0, -0.77567951].$$

The comb filter removes the 50 Hz powerline interference and its harmonics. Unlike the FIR comb filter in Section 3.2, which was designed for 50, 100 and 150 Hz, the IIR comb filter automatically produces notches at the harmonics of 50 Hz up to the Nyquist frequency. Therefore, an additional notch can be seen at 200 Hz.

The highpass filter shows a sharp feature in the group delay near 0 Hz. The comb filter also shows changes in group delay around its notch frequencies because the magnitude response becomes very small at these frequencies.

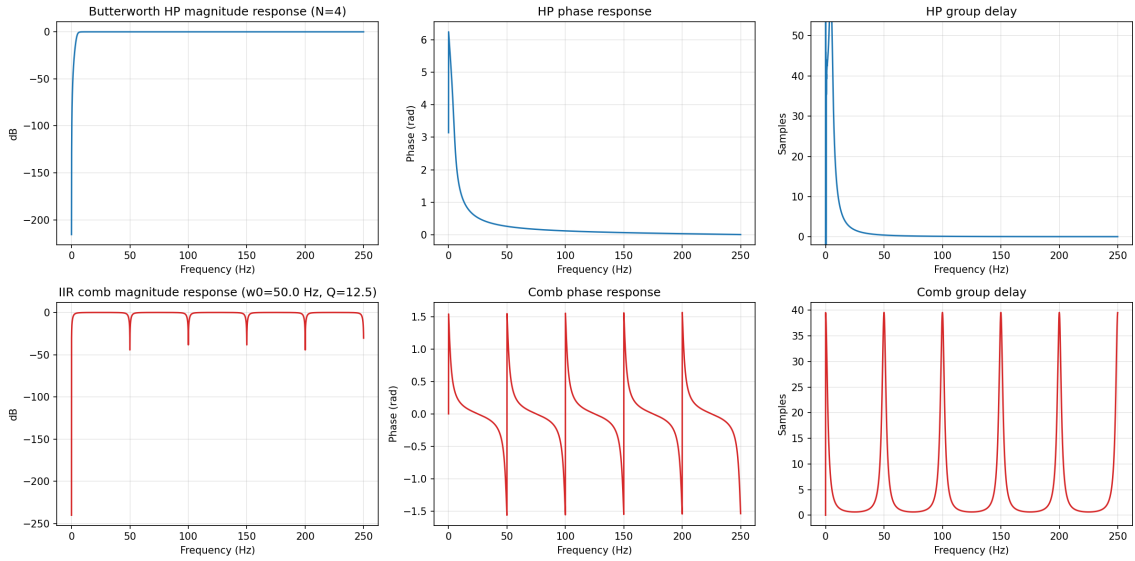


Figure 38: Butterworth highpass filter (top row) and IIR comb filter (bottom row): magnitude, phase and group delay responses.

4.1.3 Combined Response: FIR vs. IIR

Figure 39 compares the combined magnitude responses of the FIR and IIR filters. The FIR filter has deeper notches at 50, 100 and 150 Hz, while the IIR comb filter has shallower notches. This is because the FIR filter was designed with many taps to meet the required attenuation, whereas the IIR filters use a fixed order of 4 and a relatively low-order comb filter.

The IIR lowpass response decreases smoothly after 125 Hz, while the FIR response contains more oscillations in the stopband. Overall, the FIR filter provides sharper and

deeper rejection, but requires many more coefficients. The IIR filter achieves a similar overall filtering shape with a much lower order, but has a nonlinear phase response and therefore can introduce more waveform distortion when used in a single forward pass.

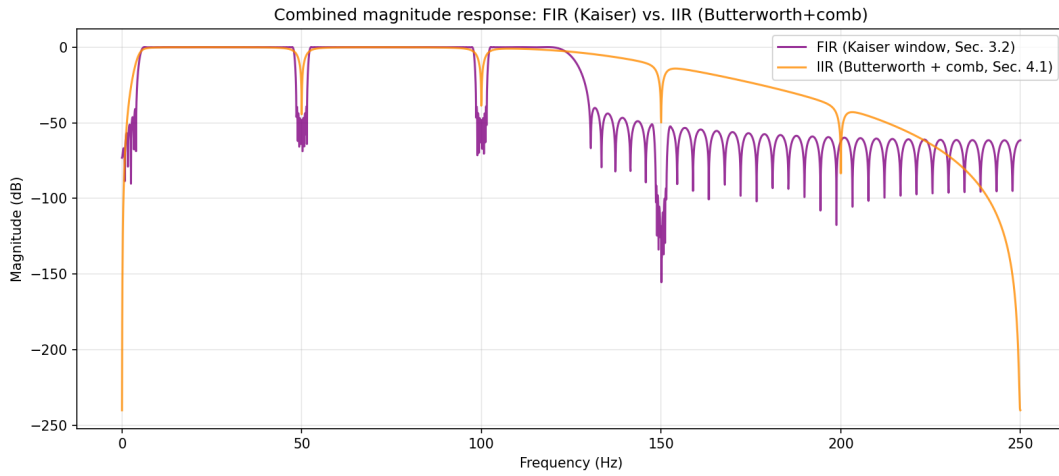


Figure 39: Combined magnitude response: FIR (Kaiser window, Section 3.2) vs. cascaded IIR (Butterworth highpass + lowpass + comb, Section 4.1).

4.2. Filtering Methods for IIR Filters

4.2.1 Forward vs. Forward-Backward Filtering: Time Domain

Figure 40 compares two ECG beats after FIR filtering, forward IIR filtering using `lfilter()`, and forward-backward IIR filtering using `filtfilt()`.

The FIR-filtered signal and the forward-backward IIR signal are visually very similar. In contrast, the forward-only IIR signal shows a time shift and a different QRS shape. This occurs because the IIR filter has a nonlinear phase response and a frequency-dependent group delay. Different frequency components of the QRS complex are therefore delayed by different amounts.

The `filtfilt()` method filters the signal in the forward direction and then in the reverse direction. The phase shift introduced by the first pass is approximately cancelled by the reverse pass, giving a near-zero-phase result. Therefore, the forward-backward IIR signal does not have the same time shift seen in the forward-only result.

4.2.2 Forward vs. Forward-Backward Filtering: PSD

Figure 41 compares the PSDs of the FIR-filtered, forward IIR-filtered, and forward-backward IIR-filtered signals. Since PSD mainly shows the signal power at each frequency, the phase differences between the filters are not visible in this plot.

All three signals have a similar passband between approximately 5 and 125 Hz, with attenuation around the powerline frequencies. Above approximately 130 Hz, the forward-only IIR signal has more remaining high-frequency energy than the other two signals. The forward-backward IIR result has stronger attenuation in the stopband. This is because `filtfilt()` applies the filter twice. The magnitude response is therefore approximately squared, which increases the attenuation in dB. As a result, the forward-backward IIR filter can provide stronger stopband attenuation than the single-pass IIR filter.

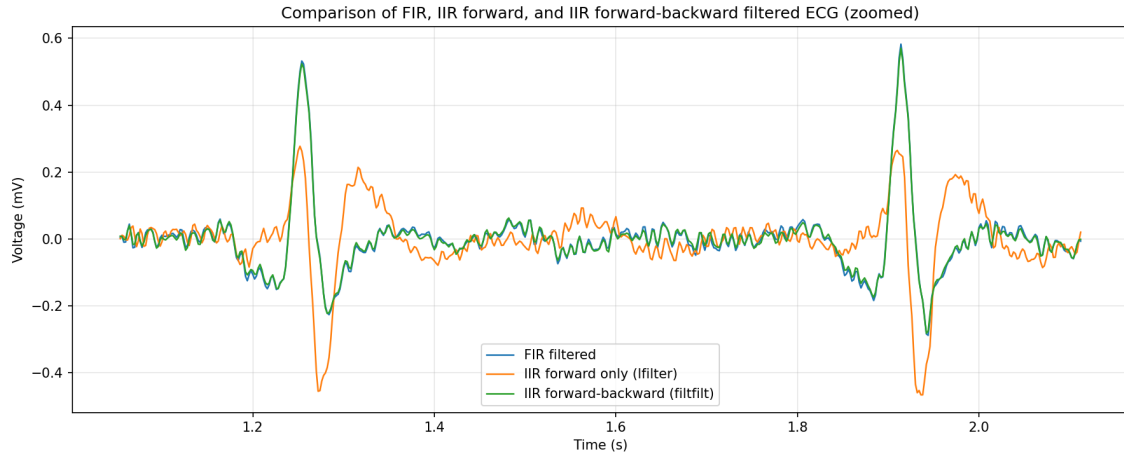


Figure 40: Zoomed time-domain comparison of FIR-filtered, IIR forward-only (`lfilter`), and IIR forward-backward (`filtfilt`) filtered ECG.

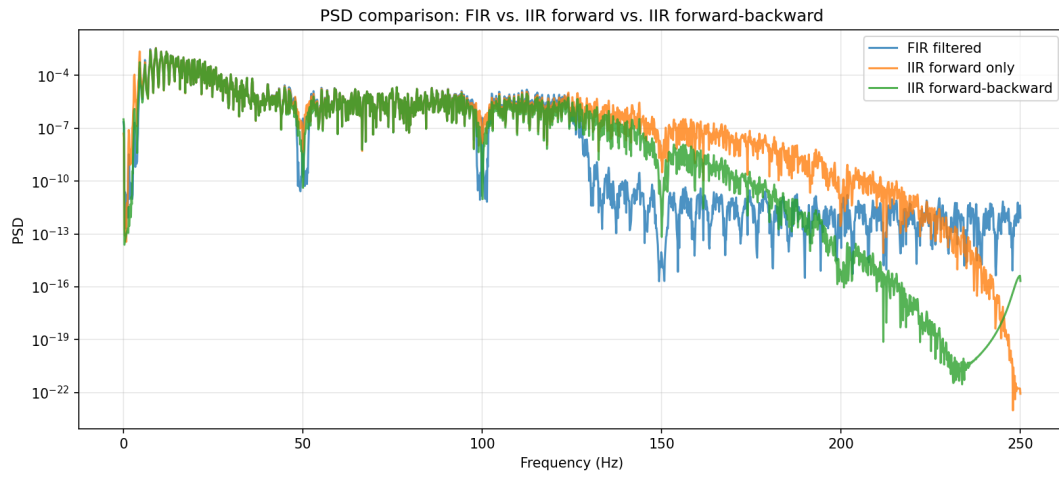


Figure 41: PSD comparison: FIR-filtered, IIR forward-only, and IIR forward-backward filtered ECG.

Overall, the time-domain results show the effect of the IIR nonlinear phase, while the PSD results show the difference in the magnitude response and attenuation of the filtering methods.